

```

clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;      % Passband cutoff (Hz)
fs = 4000;      % Sampling frequency (Hz)
N = 50;         % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');

```

--- Rectangular Window Coefficients ---

```
disp(b_rect');
```

```

0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
0
-0.0286
0
0.0349
0
-0.0449
0
0.0629
0
-0.1048
0
0.3145
0.4940
0.3145
0
-0.1048
0
0.0629
0
-0.0449
0
0.0349
0
-0.0286
0
0.0242
0
-0.0210
0

```

```
0.0185
0
-0.0166
0
0.0150
0
-0.0137
0
0.0126
```

```
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
```

```
--- Hamming Window Coefficients ---
```

```
disp(b_hamm');
```

```
0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0
-0.0379
0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
```

```

0
-0.0034
0
0.0021
0
-0.0013
0
0.0010

```

```

%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');

```

```

--- Hanning Window Coefficients ---

```

```

disp(b_hann');

```

```

0.0000
0
-0.0004
0
0.0013
0
-0.0028
0
0.0050
0
-0.0081
0
0.0122
0
-0.0179
0
0.0259
0
-0.0378
0
0.0580
0
-0.1027
0
0.3172
0.5000
0.3172
0
-0.1027
0
0.0580
0
-0.0378
0
0.0259
0
-0.0179
0
0.0122
0
-0.0081
0
0.0050
0

```

```

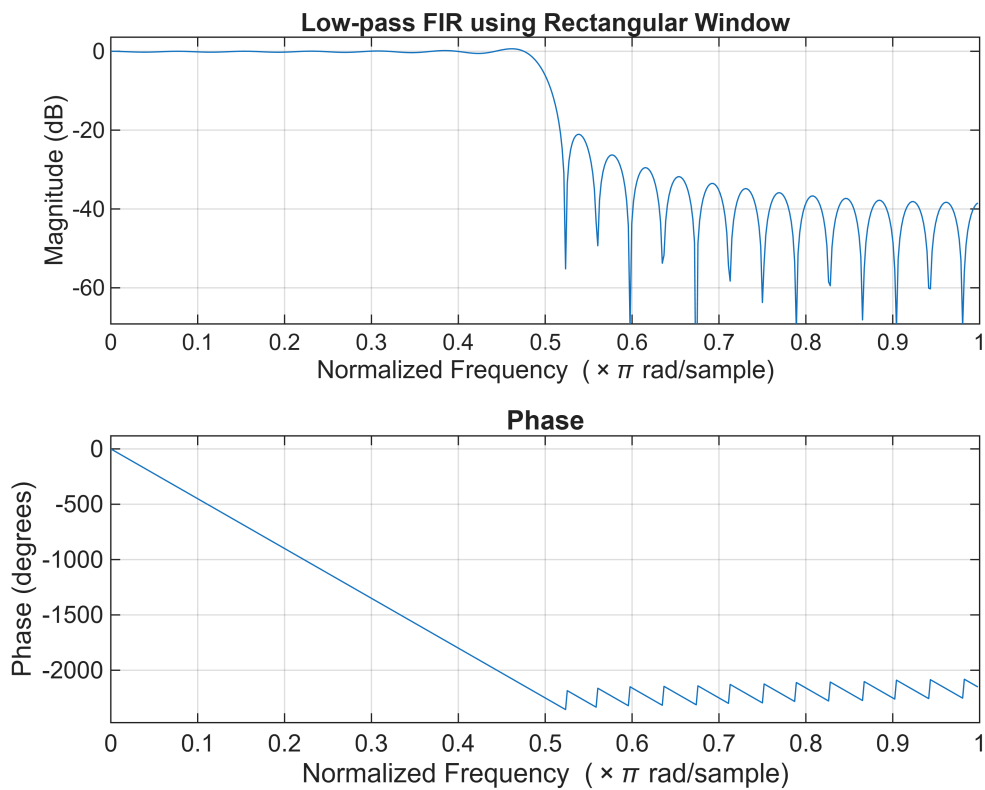
-0.0028
0
0.0013
0
-0.0004
0
0.0000

```

```

%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');

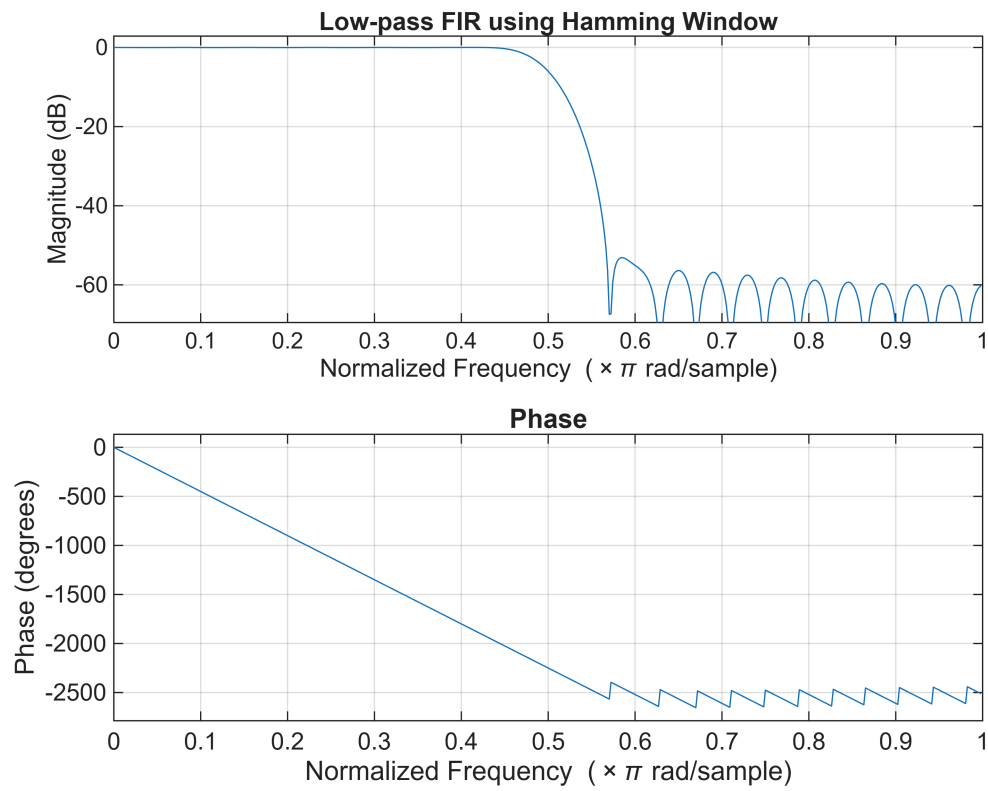
```



```

figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');

```



```
figure;  
freqz(b_hann,1);  
title('Low-pass FIR using Hanning Window');
```

